

Assignment 2 - ELG 5218 Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Problems 1.1 to 1.4 carry 5 points each (you need 15 points for the maximum grade and 5 points is the bonus). Problems 2.1 to 2.9 carry 2 points each. The maximum is 33 points, plus a 5-point bonus.

1 PROBLEM 1

- Read and understand the following: [Stochastic Variational Inference with NumPyro](#). Please note that the variational density or proposal is called the guide.
- Use the provided **student notebook** as your starting point. Complete all TODO cells and add requested plots/analysis.
- Unless stated otherwise, generate data with $N = 60$, $w_{\text{true}} = 0.5$, $b_{\text{true}} = -0.7$, $s_{\text{true}} = 1.0$, $x_i \sim \mathcal{N}(0, 5^2)$, and $y_i = w_{\text{true}}x_i + b_{\text{true}} + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, s_{\text{true}})$.
- For every training run, include: (i) the loss curve (negative ELBO), (ii) posterior summaries for parameters, and (iii) predictive checks.
- Use a fixed PRNG seed where possible so results are reproducible.
- Deliverables: one completed notebook (with outputs), and a short PDF report (1–3 pages) summarizing findings.

1.1 Polynomial regression and coefficient analysis

Consider

$$y = w_0 + w_1x + w_2x^2 + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, s),$$

with priors $w_k \sim \mathcal{N}(0, 10^2)$ and $s \sim \text{HalfCauchy}(10)$.

Start with the models and the code provided in [Stochastic Variational Inference with NumPyro](#) and modify it to do the following:

- 1.1 a** Implement the polynomial model (latent sites `w0,w1,w2,s_eps`).
- 1.1 b** Train SVI with (i) `AutoDiagonalNormal` and (ii) `AutoMultivariateNormal`. Plot both loss curves.
- 1.1 c** Plot predictive bands for both guides on the same test grid.

- 1.1 d Draw posterior samples and compute $\widehat{\text{Corr}}(w_1, w_2)$ for both guides. Explain why w_1 and w_2 can be correlated.
- 1.1 e Plot the joint posterior of (w_1, w_2) (scatter or density). Describe differences between guides.
- 1.1 f **Coefficient trajectory plot (not just ELBO):** modify your training loop to record (over epochs) the posterior mean estimates of w_0, w_1, w_2 and s (e.g., by sampling a small number of particles each checkpoint). Plot these trajectories and comment on convergence/stability.

1.2 Student- t likelihood (robust noise)

Replace the Gaussian likelihood with Student- t noise (degrees of freedom fixed at $\nu = 4$):

$$y_i | x_i, \theta \sim \text{StudentT}(\nu = 4, \text{loc} = \mu_i, \text{scale} = s).$$

- 1.2.a Implement the Student- t model and train SVI (choose a guide; start with `AutoDiagonalNormal`).
- 1.2.b Plot the loss curve and compare to the Gaussian-likelihood baseline.
- 1.2.c Plot posterior histograms for s under (i) Gaussian likelihood and (ii) Student- t likelihood.
- 1.2.d **Outlier experiment:** modify the dataset by adding 3 outliers (e.g., add +12 to three randomly chosen y_i). Refit both models and compare the posterior of s and predictive bands. Which model is more robust and why?
- 1.2.e Compute the fraction of training points covered by the 90% predictive interval for both likelihood choices (with and without outliers).

1.3 Heteroscedastic noise and learned noise curve

Let the noise depend on x :

$$s(x) = \exp(ax + c), \quad y_i \sim \mathcal{N}(wx_i + b, s(x_i)).$$

- 1.3.a Implement the heteroscedastic model with latent sites $\mathbf{w}, \mathbf{b}, \mathbf{a}, \mathbf{c}$. Train with an autoguide.
- 1.3.b Plot the loss curve and report posterior summaries for (w, b, a, c) .
- 1.3.c Using posterior samples, compute $s(x_*)$ on a dense grid and plot:
 - (i) posterior median of $s(x)$ and (ii) a 90% credible band for $s(x)$.
- 1.3.d Compare the heteroscedastic model's predictive bands with the homoscedastic baseline. In what regions of x is uncertainty larger and why?
- 1.3.e Discuss one identifiability or optimization challenge in this model (e.g., correlation between (w, b) and (a, c)).

1.4 Bonus: Hierarchical prior on w and b

To learn about Hierarchical Bayesian models please take a look at the following [video](#) and the following: [Finally! Bayesian Hierarchical Modelling at Scale](#) and [Bayesian Hierarchical Linear Regression in Numpyro](#).

Modify the model to

$$\alpha \sim \mathcal{N}(0, 10^2), \quad \tau \sim \text{HalfCauchy}(5), \quad w \sim \mathcal{N}(\alpha, \tau), \quad b \sim \mathcal{N}(\alpha, \tau),$$

with the same likelihood and noise prior for s .

1.4.a Implement the hierarchical model with latent sites `alpha`, `tau`, `w`, `b`, `s_eps`.

1.4.b Train SVI with an autoguide of your choice. (Recommended: `AutoMultivariateNormal`.) Plot the loss curve.

1.4.c Draw posterior samples and report mean/std/CI for (α, τ, w, b, s) .

1.4.d Plot marginals for α and τ . Interpret the role of τ as a shrinkage/regularization parameter.

1.4.e Compare the posterior of (w, b) under the hierarchical model to the non-hierarchical baseline (from [Stochastic Variational Inference with NumPyro](#)). What changes and why?

Submission Checklist

- Completed notebook with all outputs visible.
- PDF report with the answers to the assignment questions.

2 PROBLEM 2: Guided Study: Neural Networks and Deep Architectures

How to use this guide

This plan is designed as a **guided, active study** with checkpoints and a single assignment that follows the same structure. For each module:

- Watch the suggested videos *actively*: pause to derive/verify key equations.
- If you need additional clarification and want to learn more, please read the book chapters.
- Do the assignment and submit the answers as a pdf file. Your answers should be short - one paragraph.

Study material

Please cover the following material:

- **Training neural networks**
 - MIT 6.7960 (Lec 02. How to Train a Neural Net) https://www.youtube.com/watch?v=vidCX_dMCu0&list=PLU14u3cNGP63URZnh5iqBzDTDYPUTQT-8&index=3
 - Book chapters: [12 Neural Networks](#) and [14 Backpropagation](#)

- **CNNs**

- MIT 6.7960 (Lec 04. Architectures: Grids) <https://www.youtube.com/watch?v=bxVkJZ4M-hIE&list=PLU14u3cNGP63URZnh5iqBzDTDYPUTQT-8&index=4>
- Book chapter: [24 Convolutional Neural Nets](#)

- **RNNs**

- MIT 6.7960 (Lec 10. Architectures: Memory) <https://www.youtube.com/watch?v=IiHknRHA-Gk&list=PLU14u3cNGP63URZnh5iqBzDTDYPUTQT-8&index=10>
- : Book chapter: [25 Recurrent Neural Nets](#)

Assignment problems:

Feedforward neural networks and training basics

1. Backprop: what it computes and what it needs (one sentence + details).

- In **one sentence**, state what backpropagation computes.
- List two types of quantities that must be cached from the forward pass in order to compute gradients in the backward pass. Give one concrete example for each (e.g., ReLU, linear layer).

2. Mini-batch training and SGD as a noisy estimator.

Assume an empirical risk $J(\theta) = \frac{1}{N} \sum_{i=1}^N \ell_i(\theta)$ and mini-batch SGD uses $\hat{g}_B(\theta) = \frac{1}{B} \sum_{b=1}^B \nabla \ell_{i_b}(\theta)$ where indices are sampled i.i.d. with replacement. Assume $\text{Var}(\nabla \ell_i(\theta)) = \Sigma(\theta)$.

- Show $\mathbb{E}[\hat{g}_B(\theta)] = \nabla J(\theta)$.
- Derive $\text{Var}(\hat{g}_B(\theta))$.
- If you double B , how does the gradient noise covariance change?

3. Training curves: underfitting vs. overfitting.

You observe that *training loss decreases steadily*, but *validation loss decreases initially and then increases*.

- What phenomenon is happening? Explain briefly.
- Give three interventions, each from a different category: (i) data, (ii) regularization/model, (iii) optimization/training procedure.

4. Two signs of data leakage or evaluation bugs.

List **two** specific, concrete symptoms that suggest data leakage or a bug in evaluation (not just “accuracy is high”). For each, provide (i) what you would check first, and (ii) one quick test to confirm/refute it.

Part B: Convolutional networks (CNNs)

5. Fully-connected vs. convolution: what changes?

You replace a fully-connected layer mapping $\mathbb{R}^{CHW} \rightarrow \mathbb{R}^M$ (flattened image input) with a convolution layer mapping $x \in \mathbb{R}^{C \times H \times W} \rightarrow y \in \mathbb{R}^{M \times H \times W}$ using $K \times K$ kernels (stride 1, same padding).

- (a) Derive the parameter count of the fully-connected layer and of the convolution layer (with bias).
- (b) Explain, in 4–6 sentences, what inductive bias is introduced by convolution (locality + weight sharing).
- (c) Give one scenario where the convolutional inductive bias is inappropriate or harmful.

6. Locality/translation and pooling invariances.

- (a) Explain why CNNs exploit locality and translation structure in images (3–6 sentences).
- (b) Pooling is sometimes said to provide “invariance.” Give a precise statement: what kind of invariance (or stability) is it, and what does it not guarantee?
- (c) Give a simple counterexample showing that max-pooling with stride 2 is not exactly translation-equivariant.

7. Residual connections: why do they help? (one paragraph)

A residual block outputs $y = x + F(x; \theta)$. Write a short paragraph explaining why residual connections improve optimization for deep nets. Your answer should mention gradient flow and/or the ability to learn near-identity mappings.

Part C: Recurrent networks (RNNs) and sequence modeling

8. RNN Markov structure and what it means.

A vanilla RNN has

$$h_t = \phi(W_h h_{t-1} + U x_t + b).$$

- (a) State the conditional independence / Markov property implied by this recurrence.
- (b) Explain how the RNN can still represent long-range dependencies even though it is Markov in h_t .

9. Simple RNN forward and backward

Consider a simple RNN for a sequence of length T :

$$h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1} + b_h), \quad \hat{y}_t = W_{hy}h_t + b_y,$$

with loss $L = \sum_{t=1}^T \ell(\hat{y}_t, y_t)$.

- (a) Write the forward pass explicitly for $t = 1, \dots, T$, assuming $h_0 = 0$.
- (b) Derive the recurrence for the gradient $\partial L / \partial h_t$ in terms of $\partial L / \partial h_{t+1}$ (backpropagation through time).
- (c) Explain in 4–5 sentences how this recurrence illustrates vanishing or exploding gradients when $\|W_{hh}\|$ is much smaller or larger than 1, and relate this to difficulties in modelling long-range dependencies.